
8 Literature Themes

New Description

Outcome	Weight
Explore of IT Areas	◆◆◆◆

Outcome	Weight
Context of Research Project	◆◆◆◆◇

Outcome	Weight
Communication of Research Outcomes	◆◆◆◆

Outcome	Weight
Validation of Objectives and Methodology	◆◆◆◆◇

Outcome	Weight
Application of Skills and Knowledge	◆◆◆◆

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DEAKIN UNIVERSITY

RESEARCH TRAINING AND PROJECT

ONTRACK SUBMISSION

Literature Themes

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Research Themes and Narrative

GPS Spoofing Detection in
Autonomous Mobile Robots and UAVs

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Three Core Research Themes

GPS Vulnerabilities • Sensor Fusion Detection • Dynamic Trust Assessment
Supporting Autonomous Mobile Robot Navigation Security

Contents

1	Research Overview and Objectives	2
2	Theme 1: GPS Vulnerabilities and Platform-Specific Attack Taxonomies	2
2.1	Research Question 1: Primary GPS Vulnerabilities	2
2.2	Problem Space Definition	2
2.3	Platform-Specific Vulnerability Analysis	2
3	Theme 2: Real-Time Sensor Fusion Detection Techniques	3
3.1	Research Question 2: Sensor Fusion Implementation	3
3.2	Cross-Validation Detection Paradigm	3
3.3	Real-Time Implementation Challenges	3
3.4	Detection Methodology Framework	3
4	Theme 3: Dynamic GPS Trust Index and Reliability Assessment	4
4.1	Research Question 3: GPS Trust Quantification	4
4.2	Beyond Binary Detection	4
4.3	Trust Index Components	4
4.4	Dynamic Adaptation Mechanisms	4
5	Integrated Research Framework and Narrative Structure	5
5.1	Conceptual Integration	5
5.2	Sequential Narrative Development	5
6	Expected Contributions and Future Directions	6
6.1	Research Contributions	6
6.2	Literature Theme Associations	6
7	Conclusion	6

1 Research Overview and Objectives

Research Focus: This narrative outlines three interconnected research themes that form the foundation for investigating **GPS spoofing detection** in autonomous mobile robots through **sensor fusion techniques** and **dynamic trust assessment**.

The increasing deployment of autonomous mobile robots in critical applications—from warehouse automation to agricultural systems—has heightened the importance of navigation security. GPS spoofing attacks represent a significant cybersecurity threat that can compromise operational integrity and safety. This research narrative positions three complementary themes that together provide a comprehensive foundation for developing practical detection mechanisms.

2 Theme 1: GPS Vulnerabilities and Platform-Specific Attack Taxonomies

2.1 Research Question 1: Primary GPS Vulnerabilities

What are the primary vulnerabilities of GPS in autonomous mobile robots to spoofing attacks?

2.2 Problem Space Definition

Understanding the specific **vulnerabilities** of civilian GPS systems forms the foundational knowledge base. Current literature identifies fundamental weaknesses in civilian GPS signals but lacks comprehensive analysis specific to mobile robotic platforms. The inherently open, unauthenticated structure of GPS signals creates multiple attack vectors.

2.3 Platform-Specific Vulnerability Analysis

Research reveals distinct vulnerability patterns for mobile robots:

- **Signal reception constraints:** Ground-level operation with potential signal obstruction
- **Resource limitations:** Computational and power constraints affecting detection capabilities
- **Operating environment:** Indoor-outdoor transitions creating vulnerability windows
- **Antenna characteristics:** Limited diversity compared to aviation systems

Research Gap

Research Gap: Limited characterization of platform-specific vulnerabilities in resource-constrained mobile robots compared to aviation systems. Understanding these vulnerabilities is essential for developing targeted detection mechanisms.

Narrative Positioning

This theme defines the **problem space** specific to mobile robotics, establishing why conventional GPS security approaches may be insufficient for autonomous ground vehicles and motivating platform-specific detection strategies.

3

Theme 2: Real-Time Sensor Fusion Detection Techniques

3.1 Research Question 2: Sensor Fusion Implementation

How can sensor fusion techniques effectively detect GPS spoofing in real-time?

3.2 Cross-Validation Detection Paradigm

Recent research emphasizes **sensor fusion strategies** that cross-validate GPS data with inertial measurement units (IMU) and odometry sensors. These approaches exploit the physical independence of different sensor modalities to detect inconsistencies indicative of spoofing attacks.

3.3 Real-Time Implementation Challenges

Effective real-time detection requires addressing several technical considerations:

- **Computational efficiency:** Algorithms suitable for embedded processors
- **Sensor calibration:** Maintaining accuracy across operational conditions
- **Latency constraints:** Detection within navigation control loop timing
- **Environmental robustness:** Performance across diverse operating scenarios

3.4 Detection Methodology Framework

The core detection approach involves:

1. **Multi-sensor data acquisition:** GPS, IMU, wheel odometry integration
2. **Cross-validation analysis:** Statistical comparison of position estimates
3. **Threshold-based detection:** Dynamic adjustment based on operating conditions

4. Alert generation: Real-time notification of potential spoofing events

Research Gap

Research Gap: Insufficient real-time validation of sensor fusion methods in practical robotic navigation scenarios. Existing approaches show promise but lack implementation specifics for real-time operation on embedded platforms.

Narrative Positioning

This theme introduces **practical sensor fusion** as the primary detection paradigm, emphasizing real-time constraints and resource limitations specific to mobile robotic platforms.

4

Theme 3: Dynamic GPS Trust Index and Reliability Assessment

4.1 Research Question 3: GPS Trust Quantification

How can a dynamic GPS Trust Index be designed to quantify the reliability of GPS data?

4.2 Beyond Binary Detection

Current detection methods provide binary outputs (spoofed/authentic) but lack quantitative reliability assessment. A **dynamic trust index** enables gradual response to varying threat levels, allowing autonomous systems to adapt navigation strategies based on GPS reliability confidence.

4.3 Trust Index Components

The proposed GPS Trust Index incorporates multiple factors:

- **Signal characteristics:** C/N_0 ratios, signal strength variations
- **Consistency metrics:** Cross-sensor validation results
- **Historical reliability:** Time-weighted trust accumulation
- **Environmental context:** Operating conditions and threat assessment

4.4 Dynamic Adaptation Mechanisms

The trust index provides:

- **Continuous assessment:** Real-time reliability quantification
- **Graceful degradation:** Smooth transition between navigation modes
- **Context awareness:** Adaptation to operational environments
- **Decision support:** Informed navigation strategy selection

Research Gap

Research Gap: Absence of standardized trust quantification methods for continuous GPS reliability assessment in autonomous systems. A dynamic trust index enables gradual response to varying threat levels.

Narrative Positioning

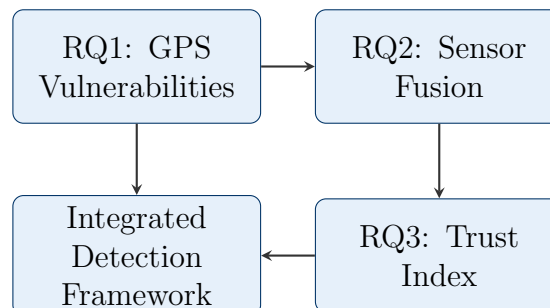
This theme advances beyond simple detection to **intelligent trust assessment**, enabling autonomous systems to make informed decisions about navigation reliability and response strategies.

5

Integrated Research Framework and Narrative Structure

5.1 Conceptual Integration

The three research themes form an integrated detection framework:



5.2 Sequential Narrative Development

The proposed narrative structure follows logical progression:

1. **Vulnerability Analysis** (Theme 1): Establishes platform-specific threat landscape
2. **Detection Implementation** (Theme 2): Practical sensor fusion for real-time operation
3. **Trust Assessment** (Theme 3): Dynamic reliability quantification and response

6

Expected Contributions and Future Directions

6.1 Research Contributions

This research will contribute to the field by:

Narrative Positioning

1. **Comprehensive vulnerability analysis** specific to mobile robotic platforms
2. **Practical sensor cross-validation algorithms** for real-time implementation
3. **Novel GPS Trust Index** for dynamic reliability assessment
4. **Reproducible simulation framework** using ROS-Gazebo-PX4

6.2 Literature Theme Associations

The research questions align with established literature themes:

- **RQ1** aligns with GPS Vulnerabilities and Attack Taxonomies
- **RQ2** corresponds to Detection Techniques and Simulation Frameworks
- **RQ3** relates to Mitigation and Recovery Strategies

7

Conclusion

This narrative framework demonstrates comprehensive understanding of GPS security challenges specific to autonomous mobile robots while constructing a technically sound foundation for the proposed minor thesis research. The three-theme structure provides clear positioning within current research trends while identifying specific contribution opportunities in robotic navigation security.

The integrated approach combining vulnerability analysis, practical sensor fusion, and dynamic trust assessment addresses current gaps in the literature and provides a pathway for developing robust, real-time GPS spoofing detection systems for autonomous mobile robots operating in diverse environments.